Transients on Transmission Lines – Part 1

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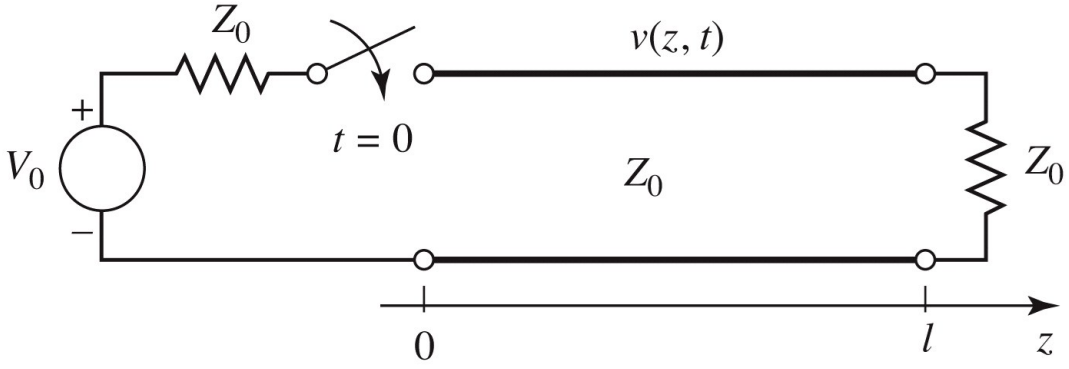
Lecture Objective

- By the end of this lecture, you will be able to:
  - Analyze the time-domain response for a transmission line terminated in an arbitrary resistive load.

Time-Domain Viewpoint

- Previously, we considered the behavior of transmission lines in the frequency domain (time-harmonic excitation).
- Frequency-domain (AC) analysis is very useful for understanding narrow band signals such as modulated RF carriers.
- Time-domain (transient) analysis is more useful for pulses or other wideband signals.

Transient Response of a Transmission Line Terminated in a Matched Load



(a)

Figure 2.21a

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2. Transient Response of a Transmission Line Terminated in a Matched Load

- At time $t = 0$, a disturbance (the leading edge of the pulse) is excited on the transmission line.
- The amplitude of the disturbance is given by voltage division between the generator impedance and the characteristic impedance of the line:

$$V^+ = \frac{Z_0}{Z_0 + Z_0} V_0 = \frac{V_0}{2}$$

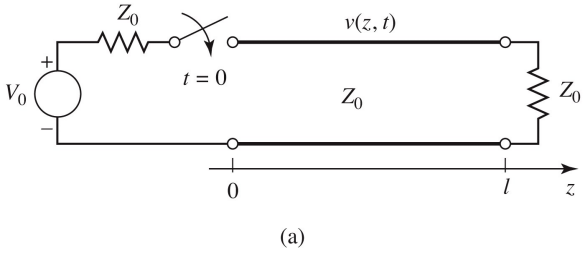

Figure 2.21a
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- This disturbance propagates down the transmission line in the +z-direction with velocity v_p and is absorbed without reflection by the load.

3. Transient Response of a Transmission Line Terminated in a Matched Load

- Thus, the voltage wave on the transmission line is given by

$$v(z,t) = V^{+} u\left(t - z/v_p\right)$$

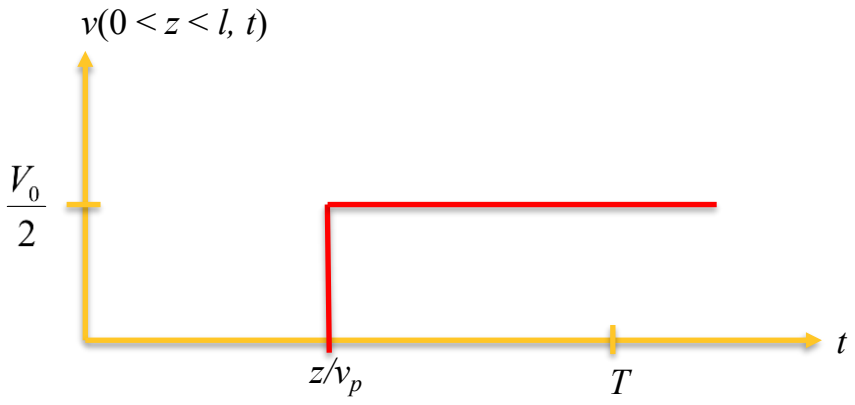
- Where $u(t)$ is the unit step function.

4. Transient Response of a Transmission Line Terminated in a Matched Load

- At time $t = T$, the disturbance is absorbed by the load, where T is the time required for the leading edge of the disturbance to traverse the entire length of the transmission line (that is to reach the load):

$$T = \frac{l}{v_p}$$

5. Transient Response of a Transmission Line Terminated in a Matched Load



6. Transient Response of a Transmission Line Terminated in a Matched Load

- So … the voltage at the load is given by

$$v_L(t) = v(l,t) = V^+ u(t-T)$$

- (DC) Steady-state solution for the load voltage:

$$V_L = v_L(t \to \infty) = \frac{V_0}{2}$$

Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

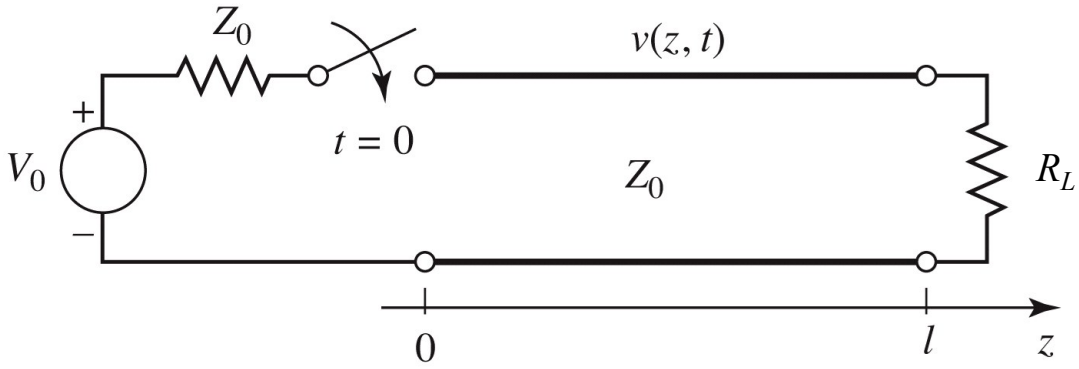


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Figure 2.21a

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2. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- At time $t = 0$, a disturbance (the leading edge of the pulse) is excited on the transmission line.
- The amplitude of the disturbance is given by voltage division between the generator impedance and the characteristic impedance of the line:

$$V^{+} = \frac{Z_0}{Z_0 + Z_0} V_0 = \frac{V_0}{2}$$

- This disturbance (incident wave) propagates down the transmission line in the +z-direction with velocity v_p until it encounters the load where some/all/none of it is reflected.

3. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- Thus, for $t < T$ (before the disturbance reaches the load), the voltage wave on the transmission line is given by

$$v(z, t < T) = V^{+} u\left(t - z/v_p\right)$$

4. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- For $t > T$ (after the disturbance reaches the load), we must have that

$$v(l, t > T) = R_L \, i(l, t > T)$$

- We have

$$v(l, t > T) = V^+ + V^-$$

$$i(l, t > T) = \frac{1}{Z_0}\left(V^+ - V^-\right)$$

5. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- Solving for the voltage reflection coefficient:

$$V^+ + V^- = \frac{R_L}{Z_0}\left(V^+ - V^-\right)$$

$$\Gamma_L = \frac{V^-}{V^+} = \frac{R_L - Z_0}{R_L + Z_0}$$

6. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- At time $t = T$, the disturbance (the leading edge of the pulse) is reflected from the load and starts traveling back toward the generator.
- The amplitude of the disturbance is given by

$$V^- = \Gamma_L V^+ = \left(\frac{R_L - Z_0}{R_L + Z_0} \right) \frac{V_0}{2}$$

- This disturbance propagates down the transmission line in the +z-direction with velocity v_p and is absorbed without reflection by the source.

7. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- Thus, the voltage on the transmission line is given by

$$v(z,t) = V^{+}u\left(t - z/v_p\right) + V^{-}u\left(t - T - \frac{l-z}{v_p}\right)$$

$$= \frac{V_0}{2}\left\{u\left(t - z/v_p\right) + \Gamma_L u\left(t - T - \frac{l-z}{v_p}\right)\right\}$$

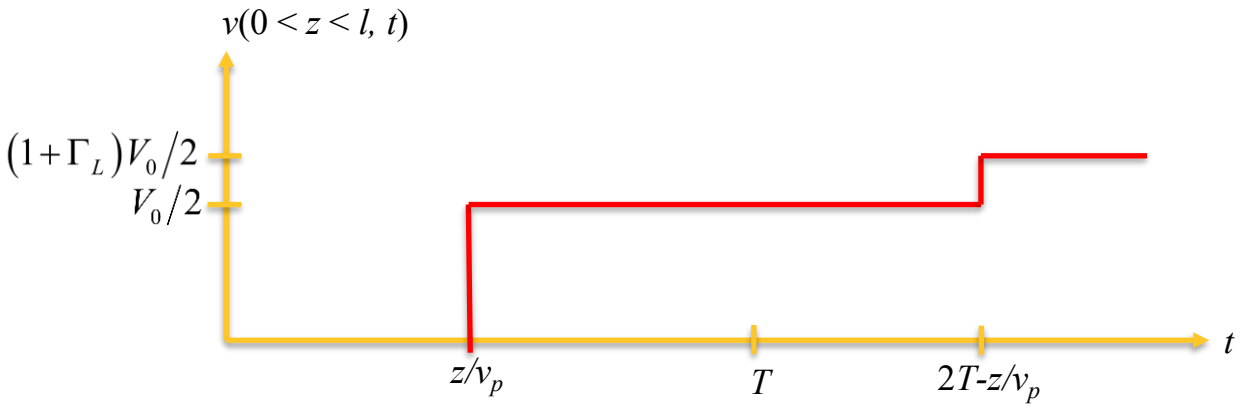
8. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load

- The current on the transmission line is given by

$$i(z,t) = \frac{V^+}{Z_0} u\left(t - z/v_p\right) - \frac{V^-}{Z_0} u\left(t - T - \frac{l-z}{v_p}\right)$$

$$= \frac{V_0}{2Z_0}\left\{u\left(t - z/v_p\right) - \Gamma_L u\left(t - T - \frac{l-z}{v_p}\right)\right\}$$

9. Transient Response of a Transmission Line Terminated in an Arbitrary Resistive Load



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